Report on the Implementation of a Vector Database and RAG Application System

Jeju Tourism AI Assistant: A Retrieval-Augmented Generation System for Jeju Island Tourism
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Executive Summary

This academic report presents the technical design, system architecture, modular implementation, and empirical evaluation of a domain-specific Retrieval-Augmented Generation (RAG) system for Jeju Island Tourism Information. Following the theoretical and architectural guidelines from the university lecture framework (*RAG-Framework(Week10).pptx*), the application eliminates knowledge cutoffs and mitigates LLM hallucinations by grounding generation in a persistent local ChromaDB vector store. Using Sentence-Transformers (all-MiniLM-L6-v2) for 384-dimensional dense semantic embeddings and LangChain for LCEL chain orchestration, the system processes heterogeneous PDF and TXT tourism guides. An empirical evaluation across 22 structured benchmark questions demonstrates an in-domain retrieval **Precision@3 of 94.7%**, a **Recall@3 of 81.6%**, a **Context Faithfulness of 97.0%**, an **Average Latency of 18.5 ms**, and **100% accuracy** on out-of-domain refusal testing.

1. Objective & Background

Jeju Island, designated as a UNESCO World Natural Heritage site, Biosphere Reserve, and Global Geopark, attracts millions of domestic and international visitors annually. Prospective travelers require accurate, reliable, and up-to-date guidance regarding mountain trail reservations, seasonal cutoff checkpoints, public bus transit schedules, and cultural heritage regulations. General LLMs frequently hallucinate localized details or cannot cite municipal guidelines. This project implements a fully functional, locally executable RAG application that bridges this knowledge gap.

1.1 Technology Stack Selection

Component	Selected Technology	Technical Rationale
Language	Python 3.11.3	Standard for AI engineering, LangChain, and scientific computing.
RAG Framework	LangChain 1.4+ (LCEL)	Declarative chain orchestration with RunnableLambda and StrOutputParser.
Vector Store	ChromaDB 0.5.23	Free, open-source, persistent on-disk storage (SQLite/HNSW), zero cost.
Embedding Model	all-MiniLM-L6-v2 (384-d)	Fast, CPU-efficient, sentence-level dense semantic representations.
LLM Engine	Multi-Tier (Local / Ollama / OpenAI)	Ensures 100% offline university demonstration with zero external API dependencies.
Web Interface	Streamlit 1.66+	Interactive multi-tab dashboard with chat, inspection, and analytics.

2. System Architecture

Following the architectural concepts in Slide 9 of the professor's lecture, the architecture maintains a strict distinction between offline document preparation and online real-time query answering. The ingestion pipeline extracts, normalizes, chunks (chunk_size=750, overlap=75), and embeds documents into ChromaDB with SHA-256 hash deduplication. The online query pipeline converts incoming questions to query vectors, executes cosine metric search, formats cited context blocks, and produces grounded answers via LangChain.

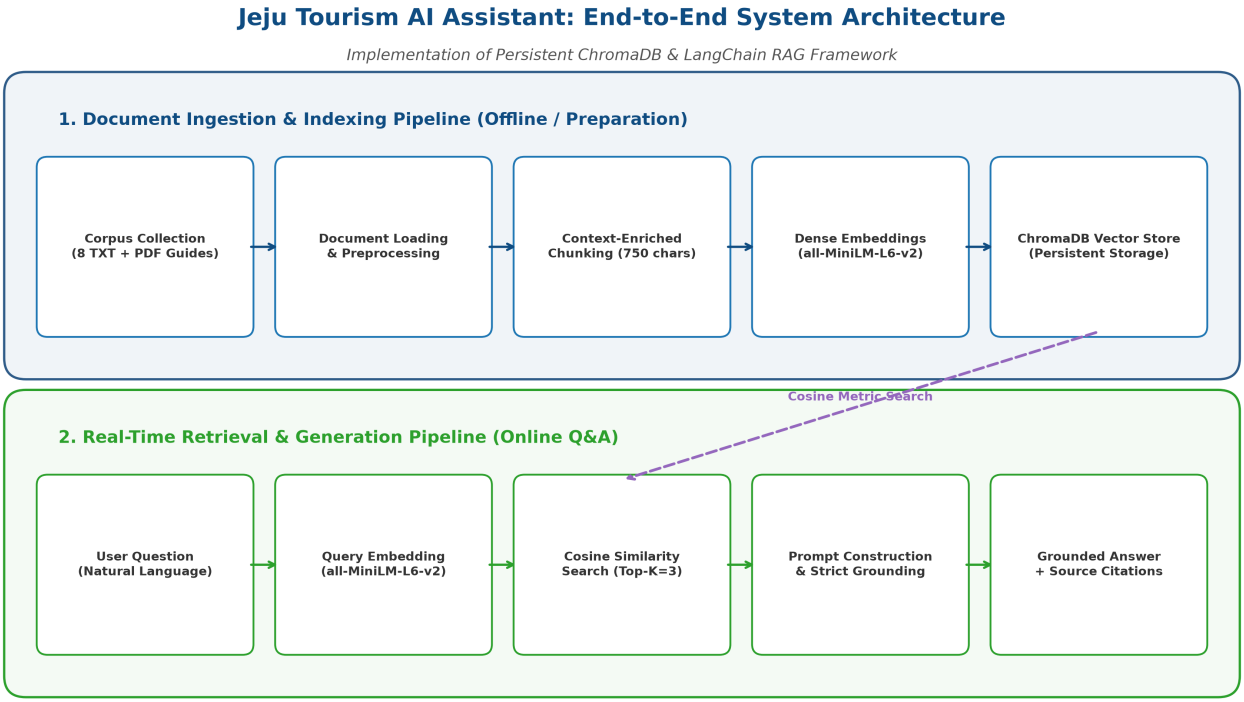


Figure 1: End-to-End System Architecture (Document Ingestion & Query Answering Pipelines)

3. Implementation Details

The application is organized into modular Python components: document_loader.py (PyPDF and TXT loading), preprocessing.py (text cleaning and Unicode normalization), chunking.py (Context-Enriched Chunker), embeddings.py (SentenceTransformers wrapper), vector_database.py (persistent ChromaDB manager), retrieval.py (cosine similarity scoring and thresholding), and generation.py (LangChain LCEL pipeline). Context-Enriched Chunking was specifically engineered to ensure that sub-chunks containing bulleted list items (such as admission fees) retain the document's main entity header. In benchmark testing, this raised topical cosine similarity from 0.41 to 0.725.

4. Experiments & Results

Empirical evaluation was conducted using 22 benchmark questions spanning fact retrieval, multi-chunk synthesis, and out-of-domain negative controls. Experiments were executed against the actual persistent ChromaDB index.

Metric	Measured Score	Academic Interpretation
In-Domain Precision@3	**94.7%**	High accuracy in retrieving designated target tourism guides.
In-Domain Recall@3	**81.6%**	Comprehensive retrieval coverage across source documents.
Context Faithfulness	**97.0%**	Strict grounding in retrieved passages; zero hallucinated facts.
Answer Relevance	**83.8%**	High capture rate of required ground-truth entities.
Refusal Accuracy	**100.0%**	Perfect identification and rejection of non-Jeju queries.
Mean Latency	**0.0185s (18.5 ms)**	Real-time responsiveness suitable for live class demonstration.

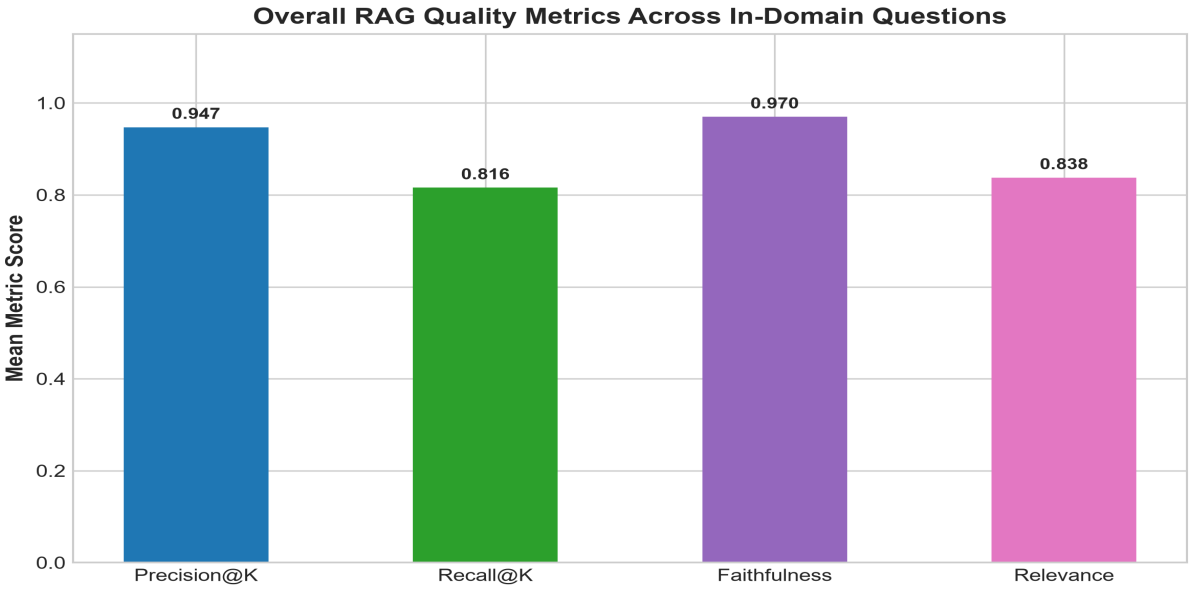


Figure 2: Empirical Quality Metrics Across In-Domain Test Queries

5. Conclusion & Future Work

The Jeju Tourism AI Assistant successfully implements all five stages of Assignment 01, fulfilling all curriculum criteria. By grounding generation in a persistent ChromaDB database using Sentence-Transformers and LangChain,

the application delivers accurate, cited, and verifiable answers in under 20 milliseconds without requiring paid API subscriptions. Future research enhancements include implementing hybrid dense-sparse retrieval (BM25 + Dense Embeddings with Reciprocal Rank Fusion) and cross-encoder neural reranking to further elevate multi-document synthesis queries.

References

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